Formalisms Every Computer Scientist Should Know: Homework Solutions

1 Homework 1

Theorem 1.1. *If there is a one-to-one function on a set A to a subset of a set B and there is also a one-to-one function on B to a subset of A, then A and B are equipollent.*

Proof. We need to show that there exists a bijection between A and B.

Assume such two functions exist. Fix the two functions $\widehat{f} : A \to B$ and $\widehat{g} : B \to A$ be one-to-one functions.

Then there exist bijections $\widehat{f}^{-1} : \widehat{f}(A) \to A$ and $\widehat{g}^{-1} : g(B) \to B$.

Define $A_C \subseteq A$ as the set of such elements $a \in A$ that $(\widehat{g} \circ \widehat{f})^n a = a$ for some integer $n \in \mathbb{N}$ (zero is not in $\mathbb{N}$).

In the same way, define $B_C = \{b \in B \mid \exists n \in \mathbb{N} : (\widehat{f} \circ \widehat{g})^n b = b\}$.

For a given element $a \in A$, define a sequence of ancestors $X(a) = (a, \widehat{g}^{-1}(a), \widehat{f}^{-1}(\widehat{g}^{-1}(a)), \ldots)$. The sequence is finite if the last element is in $A \backslash \widehat{g}(B)$ or in $B \backslash \widehat{f}(A)$, and infinite otherwise. In the same way, for $b \in B$, define $X(b) = (b, \widehat{f}^{-1}(b), g^{-1}(\widehat{f}^{-1}(b)), \ldots)$.

Define $A_I = \{a \in A \mid X(a) \text{ is infinite}\}$, $A_E = \{a \in A \mid |X(a)| \text{ is even}\}$, $A_O = \{a \in A \mid |X(a)| \text{ is odd}\}$. Observe that A_I, A_E, A_O are disjoint and cover A. Similarly, define B_I, B_E, B_O.

Notice that $A_C \subseteq A_I$, since for any $a \in A_C$, $X(a)$ is periodic and infinite.

I claim that $\widehat{f}$ maps A_I bijectively onto B_I. To see this, we need to show that for any $b \in B_I$, there exists a preimage $a \in A_I$ such that $\widehat{f}(a) = b$. Such a is the second element of $X(b)$, $a = \widehat{f}^{-1}(b)$. Then $\widehat{f}(a) = b$. a is indeed in A_I, because $X(a) = (a = \widehat{f}^{-1}(b), \widehat{g}^{-1}(\widehat{f}^{-1}(b)), \ldots)$ is also infinite.

Next, show that f maps A_E bijectively onto B_O. To see this, we need to show that for any $b \in B_O$, there exists a preimage $a \in A_E$ such that $\widehat{f}(a) = b$. Such a is the second element of $X(b)$, $a = \widehat{f}^{-1}(b)$. Then $\widehat{f}(a) = b$. a is indeed in A_E, because $X(a) = (a = \widehat{f}^{-1}(b), \widehat{g}^{-1}(\widehat{f}^{-1}(b)), \ldots)$. Then $|X(a)| = |X(b)| - 1$ is even.

In the same way, g maps B_E bijectively onto A_O. Therefore, g^{-1} maps A_O bijectively onto B_E.

Define a map $h : A \to B$:

$$h(a) = \begin{cases} \widehat{f}(a), & a \in A_I \\ \widehat{f}(a), & a \in A_E \\ \widehat{g}^{-1}(a), & a \in A_O \end{cases}, \tag{1}$$

which is the desired bijection between A and B. $\square$

2 Homework 2

Theorem 2.1 (Knaster-Tarski)**.** *Let $(A, \sqsubseteq)$ be a complete lattice and let F be a monotone function on A. Further, let $\hat{y} = \bigsqcup\{x \mid x \sqsubseteq F(x)\}$ and $\hat{z} = \bigsqcap\{x \mid F(x) \sqsubseteq x\}$. It holds, that:*

1. *$\hat{y}$ and $\hat{z}$ are fixpoints of F,*
2. *for all fixpoints x of F, $\hat{z} \sqsubseteq x \sqsubseteq \hat{y}$*

Proof. Suppose $(A, \sqsubseteq)$ is a complete lattice and let F be a monotone function on A. Let U be the set of elements $x \in A$ for which $x \sqsubseteq F(x)$ and let D be the set of elements $x \in A$ for which $x \sqsupseteq F(x)$. Then as described in the lecture let us denote the join and meet of these two sets respectively as $\hat{y} = \bigsqcup U$ and $\hat{z} = \bigsqcap D$ (this meet and join exist because the considered lattice is complete). We shall prove:

1. $\hat{y}$ and $\hat{z}$ are fixpoints of F.
2. For all fixpoints x of F the following holds $\hat{z} \sqsubseteq x \sqsubseteq \hat{y}$.

1. Let us first prove that $\hat{y}$ is a fixpoint of F. Let us pick an arbitrary element $u \in U$, then by definition of $\hat{y}$ we have $u \sqsubseteq \hat{y}$. Because F is monotone this implies $F(u) \sqsubseteq F(\hat{y})$. And because u was picked arbitrarily from U this means that $F(\hat{y})$ is an upper bound on U. But $\hat{y}$ is the least upper bound of U and thus $\hat{y} \sqsubseteq f(\hat{y})$. Again by applying the monotonicity of F we get $F(\hat{y}) \sqsubseteq F(F(\hat{y}))$ and we can conclude $F(\hat{y}) \in U$. But because $\hat{y}$ is the join of U and $F(\hat{y})$ is an element of U it follows $F(\hat{y}) \sqsubseteq \hat{y}$. Together with the inequality $\hat{y} \sqsubseteq F(\hat{y})$ this gives us $F(\hat{y}) = \hat{y}$, hence $\hat{y}$ is a fixpoint of F. The proof that $\hat{z}$ is a fixpoint of F can be done along the same lines by considering $\sqsupseteq$ instead of $\sqsubseteq$ and the set D instead of U.

2. Suppose $\hat{x}$ is a fixpoint of F. Then clearly $\hat{x} \in U$ and hence $\hat{x} \sqsubseteq \hat{y}$ as $\hat{y}$ is the join of U. Similarly $\hat{x} \in D$ and hence $\hat{x} \sqsupseteq \hat{z}$ as $\hat{z}$ is the meet of D. So it follows that $\hat{z} \sqsubseteq \hat{x} \sqsubseteq \hat{y}$ and the proof is done. $\square$

3 Homework 4

3.1 Sorting

Definition 3.1 ($\texttt{ordered}$)**.** A function $\texttt{ordered} : \Sigma^* \to \{true, false\}$ is defined as follows. $\texttt{ordered}(\varepsilon) = true$, $\texttt{ordered}(a) = true$, $\texttt{ordered}(abx) = (a \leq b) \wedge \texttt{sorted}(bx)$, where $a \in \Sigma$, $x \in \Sigma^*$.

Lemma 3.1. $\texttt{sorted}(\texttt{ms}(x)) = true$.

Proof. We will use induction on the length of x. Obviously, $\texttt{sorted}(\texttt{ms}(\varepsilon)) = \texttt{sorted}(\texttt{ms}(a)) = true$.

Recall that $\texttt{ms}(x) = \texttt{merge}(\texttt{ms}(\texttt{odd}(x)), \texttt{ms}(\texttt{even}(x)))$. By the induction assumption, we know that $\texttt{ms}(\texttt{odd}(x)), \texttt{ms}(\texttt{even}(x))$ are sorted, since their length is less than the length of x. Now it is enough to show that $\texttt{sorted}(\texttt{merge}(y, z)) = true$ if y and z are sorted. Consider $\texttt{sorted}(\texttt{merge}(ay, bz))$. Without loss of generality, let $a \leq b$, so $\texttt{sorted}(\texttt{merge}(ay, bz)) = \texttt{sorted}(a\texttt{merge}(y, bz)) = true$, since a is less or equal to both b and the next element of y, and $\texttt{sorted}(\texttt{merge}(y, bz)) = true$ holds by induction. □

Definition 3.2 ($\texttt{permutation}$)**.** For all natural n, the function $\texttt{permutation} : \Sigma^n \times \Sigma^n \to \{true, false\}$ is defined as follows. $\texttt{permutation}(\varepsilon, \varepsilon) = true$, $\texttt{permutation}(a, b) = (a == b)$, $\texttt{permutation}(ax, yaz) = \texttt{permutation}(x, yz)$, and if $a \notin y$ then $\texttt{permutation}(ax, y) = false$, where $a, b \in \Sigma$, $x, y, z \in \Sigma^*$.

Lemma 3.2. $\texttt{permutation}(x, \texttt{ms}(x)) = true$.

Proof. We will use induction on the length of x. Obviously, $\texttt{permutation}(\varepsilon, \texttt{ms}(\varepsilon)) = \texttt{permutation}(a, \texttt{ms}(a)) = true$.

First, let's show that $\texttt{merge}(\texttt{x},\texttt{y})$ is a permutation of xy. Indeed, if, for instance, $a \leq b$, then $\texttt{permutation}(\texttt{merge}(ax, by), axby) = \texttt{permutation}(a\texttt{merge}(x, by), axby) = \texttt{permutation}(\texttt{merge}(x, by), xby) = true$ by induction.

Finally, $\texttt{ms}(x) = \texttt{merge}(\texttt{ms}(\texttt{odd}(x)), \texttt{ms}(\texttt{even}(x)))$. By the induction assumption, we know that $\texttt{ms}(\texttt{odd}(x)), \texttt{ms}(\texttt{even}(x))$ are permutations of $\texttt{odd}(x), \texttt{even}(x)$ respectively, since their length is less than the length of x. Moreover, as we have already shown above, $\texttt{merge}(\texttt{ms}(\texttt{odd}(x)), \texttt{ms}(\texttt{even}(x)))$ is a permutation of $\texttt{ms}(\texttt{odd}(x))\texttt{ms}(\texttt{even}(x))$, which is a permutation of $\texttt{odd}(x)\texttt{even}(x)$. Finally, it is sufficient to prove that $\texttt{odd}(x)\texttt{even}(x)$ is a permutation of x. Using induction, $\texttt{permutation}(\texttt{odd}(ax)\texttt{even}(ax), ax) = \texttt{permutation}(a\texttt{even}(x)\texttt{odd}(x), ax) = \texttt{permutation}(\texttt{even}(x)\texttt{odd}(x), x) = \texttt{permutation}(\texttt{odd}(x)\texttt{even}(x), x) = true$. Here, we used the induction hypothesis and the fact that xy is a permutation of yx (also straightforwardly proved with induction, if needed). □

3.2 The human and monkeys theorem

Let's have a relation $<$, defined as follows: $x < y$ if $\texttt{parent}(x, y)$ or there exists z such that $x < z$ and $\texttt{parent}(z, y)$. We will assume that this relation is well-founded. Also, for all x, either $\texttt{human}(x)$ or $\texttt{monkey}(x)$.

Theorem 3.3. *If $x < y$, and* $\texttt{human}(y)$*, and* $\texttt{monkey}(x)$*, then there exist z_1, z_2 such that* $\texttt{parent}(z_1, z_2)$*, and* $\texttt{human}(z_2)$*, and* $\texttt{monkey}(z_1)$*.*

Proof. Define a sequence $w_1, w_2, \ldots$ inductively like this: $w_1 = y$, w_{n+1} is such that $\texttt{parent}(w_{n+1}, w_n)$, and $x < w_{n+1}$. If $w_n = x$, then terminate the sequence. Obviously, $w_1 > w_2 > \ldots$. Since $<$ is well-founded, the sequence has to be finite: $y = w_1 > w_2 > \ldots > w_k = x$.

For the sake of contradiction, suppose, there is no such i, that $\texttt{parent}(w_{i+1}, w_i)$, and $\texttt{human}(w_i)$, and $\texttt{monkey}(w_{i+1})$. Then, by induction, all w_i are humans. This contradicts the fact that w_k is a monkey. Therefore, the desired z_1, z_2 exist among some w_{i+1}, w_i, as requested. □

4 Homework 6: Soundness of FOL Resolution

In the following we use sets for both representing formulas in CNF and individual clauses. Hence, a formula in CNF is a set containing sets of literals.

We prove the soundness of the following FOL resolution rule:

$$\frac{C_1[l_1] \qquad C_2[\neg l_2]}{C_1\theta[\bot] \vee C_2\theta[\bot]} \; l_1\theta = l_2\theta$$

Lemma 4.1. *For any $C_1[l]$, $C_2[\neg l]$ first-order clauses and $\mathcal{I}$ interpretation, we get*

$$\mathcal{I} \models \{C_1[l], C_2[\neg l]\} \Rightarrow \mathcal{I} \models C_1[\bot] \vee C_2[\bot].$$

Proof. Let $\mathcal{I}$, $C_1[l]$, $C_2[\neg l]$ be arbitrary and $\mathcal{I} \models C_1[l]$ and $\mathcal{I} \models C_2[\neg l]$. We show $C_1[\bot] \vee C_2[\bot]$. Consider the two cases for $[\![l]\!]_{\mathcal{I}}$:

Case 1. $[\![l]\!]_{\mathcal{I}} = true$ **:** Consider the clause $C_2' = C_2 \setminus \{\neg l\}$, corresponding to the clause C_2 with $\neg l$ removed. Since $[\![\neg l]\!]_{\mathcal{I}} = false$, and $\mathcal{I} \models C_2[\neg l]$, $\mathcal{I} \models C_2'$. Moreover, since $\mathcal{I} \models C_2'$, $\mathcal{I} \models C_2' \cup \{\bot\}$ and therefore $\mathcal{I} \models C_2[\bot]$. Finally, since $\mathcal{I} \models C_2[\bot]$, $\mathcal{I} \models C_1[\bot] \vee C_2[\bot]$.

Case 2. $[\![l]\!]_{\mathcal{I}} = false$ **:** symmetric to Case 1. □

Lemma 4.2. *Let $\mathcal{I}$, C be an arbitrary interpretation and clause, and σ a substitution.*

$$\mathcal{I} \models \forall x_1 ... x_n, C \Rightarrow \mathcal{I} \models \forall x_1 ... x_n, C\sigma$$

Proof. Let $\mathcal{I}, C, \sigma$ be arbitrary with $\mathcal{I} \models \forall x_1 ... x_n C$, and assume towards a contradiction that $\mathcal{I} \not\models \forall x_1 ... x_n, C\sigma$. By the definition of $[\![\forall x.\phi]\!]$, there must be a mapping $\kappa = [y_1 \to d_1, ..., y_n \to d_n]$ from free variables in $C\sigma$ to $d_1 ... d_n \in D_{\mathcal{I}}$ s.t. $[\![C\sigma]\!]_{\mathcal{I}[\kappa]} =$ false.
We now construct a mapping $\kappa' = x_1 \to d_1', ..., x_n \to d_n'$ s.t. $[\![C]\!]_{\mathcal{I}[\kappa']} = false$. Consider the syntactic substitution $\sigma = x_1 \to t_1, ..., x_n \to t_n$, we simply define $\kappa' = [x_1 \to t_1\kappa, ..., x_n \to t_n\kappa]$. Hence, κ' is a map from free variables in C to elements in the domain $D_{\mathcal{I}}$ and is a valid mapping to close the formula C.

Moreover, $[\![C]\!]_{\mathcal{I}[\kappa']} = [\![C\sigma]\!]_{\mathcal{I}[\kappa]} = false$. Hence, $\mathcal{I} \not\models \forall x_1, ..., x_n C$, a contradiction. $\square$

Theorem 4.3 (Soundness of FOL resolution)**.**
For any clause set S, clauses $C_1[l_1] \in S$, $C_2[\neg l_2] \in S$, and substitution θ s.t. $l_1\theta = l_2\theta$: S is satisfiable iff $S \cup \{C_1\theta[\bot] \vee C_2\theta[\bot]\}$ is satisfiable.

Proof.

$\Rightarrow$ Let S, $C_1[l_1] \in S$, $C_2[\neg l_2] \in S$ s.t. $l_1\theta = l_2\theta$, be arbitrary and S satisfiable. By definition of satisfiable, let $\mathcal{I}$ s.t. $\mathcal{I} \models S$ and therefore $\mathcal{I} \models C_1$, and $\mathcal{I} \models C_2$. Note, that both clauses C_1 and C_2 are implicitely universally quantified, and since $\forall$ distributes over $\wedge$, we get $\mathcal{I} \models \forall x_1...x_n, C_1$ and $\mathcal{I} \models \forall x_1...x_n, C_2$ where $x_1...x_n$ are the free variables in S.
By Lemma 4.2, $\mathcal{I} \models \forall x_1...x_n, C_1\theta$ and $\mathcal{I} \models \forall x_1...x_n, C_2\theta$. In the following we drop the implicitely quantified variables again. Let $l = l_1\theta = l_2\theta$ and since the order of substitution is irrelevant, $\mathcal{I} \models C_1[l]\theta$ and $\mathcal{I} \models C_2[l]\theta$. By Lemma 4.1, $\mathcal{I} \models C_1\theta[\bot] \vee C_2\theta[\bot]$, and therefore, by definition of satisfiable, $S \cup \{C_1\theta[\bot] \vee C_2\theta[\bot]\}$ is satisfiable.

$\Leftarrow$ Let S, $C_1[l_1] \in S$, $C_2[\neg l_2] \in S$ s.t. $l_1\theta = l_2\theta$, be arbitrary and $S \cup \{C_1\theta[\bot] \vee C_2\theta[\bot]\}$ satisifable. By definition of satisfiable, let $\mathcal{I}$ s.t. $\mathcal{I} \models S \cup \{C_1\theta[\bot] \vee C_2\theta[\bot]\}$. By definition of $\wedge$, $\mathcal{I} \models S$ and therefore S is satisfiable. $\square$

5 Homework 9

5.1 Formulate QBF as a first-order theory

We define the theory T_{QBF} over the signature $\langle\{t^0, f^0\}, \{v^1\}\rangle$ with constant function symbols t and f and a unary predicate v.

5.1.1 Define a first-order theory T_{QBF}

We axiomatize the theory:

1. $\forall x, x = t \vee x = f$
2. $t \neq f$
3. $v(t)$
4. $\neg v(f)$

The first two axioms restrict the models to exactly those models with a domain of cardinality of two. The third and fourth axiom ensure that the constant t is interpreted as true and f as false.

5.1.2 Define a function f from closed QBF formulas to T_{QBF} formulas s.t. a formula ϕ is true iff $f(\phi)$ is satisfiable.

Let $\mathcal{P}$ be the set of propositions. We define a function $f : \mathcal{P} \to \mathcal{A}$ from propositions to atomic formulas. We consider formulas as defined in the lecture and add existential quantification and $\top$ as they are used in QBF.

- $f(\top) \mapsto \top$
- $f(\bot) \mapsto \bot$
- $f(p) \mapsto v(p)$

We then get a function $\mathcal{F} : \psi_{QBF} \to \psi_{T_{QBF}}$ by mapping f over the connectives of FOL.

Lemma 5.1. *For any closed QBF formula ϕ, ϕ is true $\iff$ $\mathcal{F}(\phi)$ is satisfiable.*

Proof. We will consider the natural structure $\mathcal{I} \coloneqq \langle \{0,1\}, \{t,f\}, \{v\} \rangle$ with $t_{\mathcal{I}} = 1$, $f_{\mathcal{I}} = 0$ and $v = \{1\}$. We omit writing the subscript $\mathcal{I}$ in the following. Clearly, the structure is a T_{QBF}-*Model*. We prove that ϕ is true *iff* $\mathcal{I} \models \mathcal{F}(\phi)$ by induction on the structure of ϕ.

- $\phi \coloneqq \bot$. For both directions we get an immediate contradiction as $\mathcal{I} \not\models \bot$ and $\bot$ is not true.
- $\phi \coloneqq \top$, trivially $\mathcal{I} \models \mathcal{F}(\top) = \top$ and $\top$ is true, hence $\top$ is true *iff* $\mathcal{I} \models \mathcal{F}(\top)$.
- $\phi \coloneqq p$, (for completenes, ϕ is closed)
- $\phi \coloneqq \phi_1 \Rightarrow \phi_2$. Assume ϕ is true, then either ϕ_1 is false or both ϕ_1 and ϕ_2 are true. By induction, in the first case we get $\mathcal{I} \not\models \mathcal{F}(\phi_1)$, and therefore $\mathcal{I} \models \mathcal{F}(\phi)$ and in the second case $\mathcal{I} \models \mathcal{F}(\phi_1)$ and $\mathcal{I} \models \mathcal{F}(\phi_2)$, hence $\mathcal{I} \models \mathcal{F}(\phi)$. For the other direction, assume $\mathcal{I} \models \mathcal{F}(\phi)$. Then either $\mathcal{I} \not\models \mathcal{F}(\phi_1)$ or both $\mathcal{I} \models \mathcal{F}(\phi_1)$ and $\mathcal{I} \models \mathcal{F}(\phi_2)$. By induction, in the first case ϕ_1 is false, hence ϕ true and in the second case both ϕ_1 and ϕ_2 are true, hence ϕ is true.
- $\phi \coloneqq \forall \phi_1$. By (IH) we know that $\phi_1[\top]$ is true *iff* $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ and $\phi_1[\bot]$ is true *iff* $\mathcal{I} \models \mathcal{F}(\phi_1[\bot])$. We continute to show that $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ *iff* $\mathcal{I}_{x \leftarrow 1} \models \mathcal{F}(\phi_1)$ and $\mathcal{I} \models \mathcal{F}(\phi_1[\bot])$ *iff* $\mathcal{I}_{x \leftarrow 0} \models \mathcal{F}(\phi_1)$, hence by definition of $\forall$, $\mathcal{I} \models \mathcal{F}(\phi)$

  case $\top$**:** Since $\mathcal{I} \models \top \iff v(t)$, we know $\mathcal{I} \models \mathcal{F}(\phi_1[v(t)])$ *iff* $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ (*Replacement theorem* would require another (trivial) proof by induction on formula). Moreover, $\mathcal{I} \models \mathcal{F}(\phi_1[v(t)]) \iff \mathcal{I}_{x \leftarrow 1} \models \mathcal{F}(\phi_1[v(x)])$ and $\phi_1[v(x)] = \phi_1$. By transitivity of $\iff$, $\phi_1[\top]$ is true *iff* $\mathcal{I}_{x \leftarrow 1} \models \mathcal{F}(\phi_1)$.

case $\bot$**:** Symmetric to **case** $\top$.

- $\phi := \exists \phi_1$. By definition, either $\phi_1[\bot]$ or $\phi_1[\top]$ is true. We proceed by cases, by (IH) we know that in the first case $\phi_1[\top]$ is true *iff* $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ and in the other case $\phi_1[\bot]$ is true *iff* $\mathcal{I} \models \mathcal{F}(\phi_1[\bot])$. For each case, the proof is identical to the cases $\bot$ and $\top$ above.

To prove the final statement, assume ϕ is true, then by the proof above $\mathcal{F}(\phi)$ is satisfiable as $\mathcal{I} \models \mathcal{F}(\phi)$. For the other direction, assume $\mathcal{F}(\phi)$ is satisfiable, then there exists an $\mathcal{I}'$ s.t. $\mathcal{I}' \models \mathcal{F}(\phi)$. By the axioms, $\mathcal{I}'$ is isomorphic to $\mathcal{I}$, since $\mathcal{I}'$ is a T_{QBF}-Model and therefore by axioms (1) and (2) has a 2 element domain, with the element $x = [\![t]\!]'_{\mathcal{I}}$ being the only element s.t. $x \in [\![v]\!]'_{\mathcal{I}}$. Thereofre for any ψ, $\mathcal{I}' \models \psi$ *iff* $\mathcal{I} \models \psi$. Hence, also $\mathcal{I} \models \mathcal{F}(\phi)$ and therefore ϕ is true. $\square$

5.2 Argue that Pressburger arithmetic does not allow QE.

The goal of this exercise is to show that Presburger arithmetic $\mathcal{T}_{PR}$ as defined in the lecture (i.e., without divisibility predicate) does *not* allow for quantifier elimination (QE).

It is important to point out that in most cases, Presburger arithmetic is considered with the divisibility predicate, which allows for quantifier elimination, this construction was even given by Presburger in his original 1929 paper.

To show that QE does not work without the divisibility predicate, consider the $\mathcal{T}_{PR}$ formula φ that expresses that a variable y is even.

$$\varphi : \quad \exists x.\ x + x \le y \wedge y \le x + x$$

Note that in this formula, y is a *free variable*. Depending on the value of y, the formula φ is either true or false. The set of values of y for which φ is true is the set of even numbers, which is countably infinite.

For the sake of readability, we say a formula ψ with a single free variable x *defines* a subset of the natural numbers S if ψ evaluates to true if and only if $x \in S$. We now claim that any *quantifier-free* $\mathcal{T}_{PR}$-formula ψ with a single free variable defines a subset of the natural numbers that can be expressed as a finite union of intervals. To show this, we first consider atomic formulas and then proceed by induction on the structure of ψ.

Lemma 5.2. *Every quantifier-free, atomic $\mathcal{T}_{PR}$-formula ψ with a single free variable x defines a contiguous subset of the natural numbers.*

Proof. Denoting repeated addition of x by ax using some constant a, we can express any atomic formula in $\mathcal{T}_{PR}$ as $ax + b \le cx + d$, which simplifies to $(a - c)x \le d - b$. This atomic formula is true for integer values of x in the following intervals:

- When $a > c$:
  - If $d \ge b$, then $x \in \left[0, \frac{d-b}{a-c}\right]$.

– If $d < b$, then $x \in \left[\frac{d-b}{a-c}, \infty\right)$. However, since $\frac{d-b}{a-c}$ is negative and $x \in \mathbb{N}$, this effectively means $x \in [0, \infty)$.

- When $a < c$:
  – If $d \geq b$, then $x \in \left[\frac{d-b}{a-c}, \infty\right)$.
  – If $d < b$, then $x \in \left[0, \frac{d-b}{a-c}\right]$.
- When $a = c$:
  – If $d \geq b$, then $x \in \mathbb{N}$.
  – If $d < b$, the formula is false, i.e. x is in the empty interval.

□

Theorem 5.3. *Any quantifier-free $\mathcal{T}_{PR}$-formula ψ with a single free variable x defines a finite union of intervals.*

Proof. We proceed by induction on the structure of ψ.

Base case: By Lemma 5.2, if ψ is an atomic formula, it defines a contiguous subset of the natural numbers (i.e., a single interval). **Induction step**: Let ψ be a boolean combination of formulas φ_1 and φ_2, that both describe finite unions of intervals.

- $\psi := \neg\varphi_1$: This describes the complement of the intervals defined by φ_1. The complement of a finite union of intervals is a finite union of intervals.
- $\psi := \varphi_1 \wedge \varphi_2$: This describes the intersections of the intervals defined by φ_1 and φ_2. The intersection of two finite unions of intervals is a finite union of intervals.
- $\psi := \varphi_1 \vee \varphi_2$: This describes the union of the intervals defined by φ_1 and φ_2. The union of two finite unions of intervals is still a finite union of intervals.

□

Observe that every finite union of intervals can only describe a subset of the natural numbers of a certain structure:

- A finite set of natural numbers, if the union does not contain the interval $[k, \infty)$ for some $k \in \mathbb{Q}^+ \cup \{0\}$.
- Otherwise, the union of a finite set of natural numbers and $\{n \mid n \in \mathbb{N} \wedge n \geq k\}$ for some $k \in \mathbb{Q}^+ \cup \{0\}$.

It is easy to see that thus, every subset of natural numbers defined by a finite union of intervals is either *finite* or *has a finite complement* (when it contains $[k, \infty)$ for some k). This is in contrast to the definition of any periodic set of natural numbers like the even numbers, which is both countably infinite and has a countably infinite complement. Therefore, any periodic set of natural numbers cannot be described by a quantifier-free $\mathcal{T}_{PR}$-formula.